

CONTINGENCY PLAN FOR ADVERSE EXTREME TEMPERATURES HEAT MODULE 2014

Services Department
Disease Prevention and Health Promotion
Division of Environmental and Occupational Health

Contingency Plan for Temperatures Extreme Adverse Conditions – Heat Module 2014

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ACES	Group of Health Centers
NAPC	National Civil Protection Authority
WHAT	Portuguese Environment Agency
ARS	Regional Health Administration
CCDR	Regional Coordination and Development Commission
DEP	Department of Epidemiology
DSC	Health Delegate Coordinator
DSP	Department of Public Health
DSR	Regional Health Delegate
INEM	National Institute of Medical Emergency
BUT	National Institute of Health Dr. Ricardo Jorge
IPMA	Portuguese Institute of the Sea and the Atmosphere
ISS	Social Security Institute
ULS	Local Health Unit

1. INTRODUCTION

According to a study by the European Environment Agency, the last decade (2002-2011) was the warmest in Europe, and the average surface temperature of the atmosphere was 1.3°C warmer than the average in the pre-industrial period, with heat waves increasing in frequency and duration. Projections made by this Agency indicate that global climate change will lead to the intensification of various extreme weather phenomena, such as heat waves, which may become more intense and frequent, associated with hotter summers and milder winters, with impacts at the social, environmental and human health levels. (EEA, 2012).

Due to its geographical location, Portugal is expected to be one of the European countries most vulnerable to climate change and extreme weather events. Some studies conducted in Portugal suggest a clear trend towards an increase in average temperature and an increase in the number of days per year with high temperatures (Santos, F. D., Miranda, P., Ed., 2006).

In 2014, the Portuguese Institute of the Sea and the Atmosphere, in partnership with the Dom Luiz Institute of the University of Lisbon, completed global scenarios whose results indicate that the average warming in the territory at the end of the 21st century, for two of the socioeconomic scenarios analyzed, increases by approximately 2.5°C and 4°C, respectively. (IPMA)

Portugal has been particularly affected in recent years by extreme weather events such as the droughts of 2004/2005 and 2011/2012, and the fact that the spring of 2009 was the driest since 1931. Prolonged drought periods, coupled with a significant increase in the number of fires as a result of longer periods of high temperatures, may further contribute to the increased risk of desertification.

The heat wave that occurred in 2003 lasted for more than two weeks in some areas of the country, and was associated with an excess mortality of 1,953 deaths, with a particular incidence in individuals aged 75 years or older (DGS, INSA). In 2010, four climatological heat waves occurred between May 17 and August 11, with an estimated excess mortality of 2,167 deaths during periods of intense heat (Final Evaluation Report – PCOC 2010). For the period between June 23 and July 14, 2013 (a period that included the two main heat waves of that year), INSA determined an excess mortality of 1,684 deaths.

These changes in the frequency and intensity of extreme weather events pose serious risks to human health, with the potential to increase the number of deaths associated with intense heat, cardiorespiratory problems related to air pollution, waterborne and foodborne illnesses, as well as the occurrence of disease vectors.

The effects of high temperatures and heat waves depend on the level of exposure (frequency, severity, and duration), the size of the exposed population, and the vulnerability of the population. Therefore, it is not surprising that the relationship between temperature and its effects on health shows some heterogeneity among populations and depending on their geographical location. On the other hand, some studies indicate that the occurrence of extreme temperatures at the beginning of the summer season is usually associated with a higher number of deaths compared to later occurrences. According to the Portuguese Institute of the Sea and the Atmosphere, June is the summer month in which heat waves occur most frequently in mainland Portugal.

It is also worth highlighting the importance of the "urban heat island" phenomenon, which, by definition, occurs within cities and corresponds to an increase in the temperature of urban pavements and air relative to the immediate surroundings, occurring mainly under conditions of calm atmosphere and clear skies. Studies carried out for the main Portuguese cities indicate average heat island intensity values of 1°C to 4°C for Lisbon, around 2.8°C for Porto, and around 1.5°C for Coimbra and Évora (Alcoforado, MJ, 2009).

While its effects are felt primarily in the early evening, it can prove important during periods of intense heat, as it hinders body cooling and prolongs the negative effects of heat on human health.

The weather conditions that occur during periods of intense heat/heat waves favor the worsening of air pollution, particularly pollutants such as tropospheric ozone and particulate matter. Since high temperatures and air pollution often coincide, it can be difficult to isolate the health effects resulting from the two exposures (Euroheat, 2009).

On the other hand, in Portugal, during the spring and summer months, the presence of particles in the atmosphere originating from the arid zones of North Africa, namely the Sahara and Sahel deserts, can be observed more frequently, with most of these particles being of natural origin. These natural events may affect ambient air quality and contribute to an increase in the concentration of suspended particles and dust, potentially causing harmful effects to human health.

Currently, many European countries, including Portugal, have implemented surveillance and alert systems, which include actions and preventive measures to address health risks associated with extreme weather events. These systems constitute important adaptation measures to climate change, aiming to improve the performance of health and social response services during periods of increased risk, thus contributing to increased population resilience.

The Ministry of Health, through the DGS (Directorate-General of Health), has implemented the Heat Wave Contingency Plan since 2004, with the aim of minimizing the negative effects of heat on health. This Plan is intended to be a strategic instrument, enhancing inter-institutional coordination between the different sectors of the Central Public Administration and its decentralized services, and with the Local Administration.

Indeed, managing the risk associated with heat for public health, a problem that cuts across society, requires the mobilization not only of health structures but also of all entities responsible for protecting the population, namely the Social Security Institute and the National Civil Protection Authority, the decentralized services, and the Local Administration.

In 2006, the procedures of the Heat Wave Contingency Plan were updated by order of the Director-General of Health, particularly regarding the assessment and management of health risks to the population, assigning this responsibility to...

Competent Health Authorities, at the regional (Regional Health Administration) and local levels. (Local Health Units/Groups of Health Centers), given their knowledge of local specificities, particularly geodemographic and in terms of management of resources.

It is the responsibility of the Regional Health Delegate and the Coordinating Health Delegates to coordinate and initiate the measures deemed necessary at any given time, based on the risk assessment carried out and in accordance with the competences defined in the Heat Module of the...

Regional Contingency Plans 2014, taking into account the responsibilities outlined in Decree-Law 135/2013, of October 4, for entities exercising sanitary authority, and the provisions of Decree-Law 137/2013, of October 7, regarding the registration of the organization and operation of ACES (Local Health Units).

In 2011, due to evidence that exposure to high temperatures constitutes a risk to human health, even without a heat wave (climatological definition), the designation of Contingency Plan for Adverse Extreme Temperatures – Heat Module was adopted.

Thus, the implementation of the Contingency Plan has been a dynamic process, as it has been progressively adapted according to the needs encountered, in order to improve its effectiveness in achieving its objectives.

The Contingency Plan for Adverse Extreme Temperatures 2014 – Heat Module, includes a strategy for maximizing resources, always bearing in mind their efficient use; a prevention strategy, because it is more efficient to promote health and prevent disease than to cure patients; a strategy for coordination between the various partners mentioned above, reinforcing interministerial/intersectoral coordination; and a strategy for community participation, since populations must also be committed to this process of protecting health.

The Heat Module of the 2014 Contingency Plan is, therefore, a strategic instrument aimed at protecting the health of populations against the negative effects of periods of intense heat. To this end, this Plan is based on a system of forecasting, alerting, and appropriate response, being activated between May 15 and September 30, 2014, and may be activated, depending on the meteorological conditions, at any time of the year, before or after its activation period.

2. PHYSIOLOGICAL RESPONSE TO INTENSE HEAT

Body temperature is constantly regulated within a relatively narrow range (35°C – 39°C) so that optimal physiological function is preserved. During periods of moderate atmospheric temperatures, the normal internal temperature of the human body (approximately 36.5°C) is maintained by the hypothalamus and balanced through equal rates of heat gain and heat loss by the body.

The main source of heat gain is the body's own internal heat, called metabolic heat, which is produced by the biochemical processes that keep us alive and by the energy we use in physical activity. The human body exchanges heat with the outside mainly through the processes of radiation, convection, and evaporation through sweat (Figure No. 1).

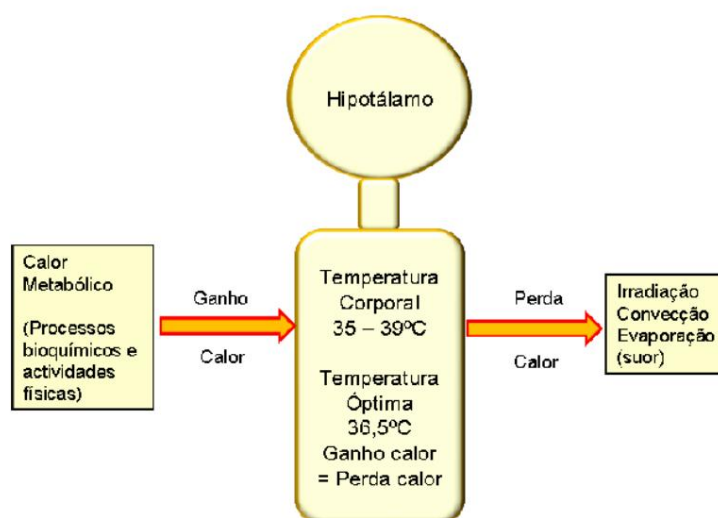


Figure No. 1 - Main forms of heat transfer between the human body and the environment.

Excessive exposure to heat is a *stress* factor for the body, particularly for the cardiovascular system.

As long as the skin temperature is higher than the ambient temperature, there will be heat loss through radiation and convection. When environmental conditions dominate the body's heat dissipation mechanisms, the internal body temperature increases.

In this case, when heat gain is greater than heat loss, the body's internal temperature increases beyond what is considered normal, and in extreme situations it can pose a high health risk ($>40^{\circ}\text{C}$).

Variations, usually of less than 1°C , occur with the time of day, level of physical activity, or emotional state. A change in body temperature exceeding 1°C occurs only during illness or when environmental conditions exceed the body's ability to cope with extreme temperatures.

An increase of less than 1°C is immediately detected by thermoreceptors scattered throughout the skin, tissues, and deep organs. The thermoreceptors transmit the information to the thermoregulatory center in the hypothalamus, which triggers two powerful responses aimed at increasing heat dissipation.

The two basic physiological mechanisms by which the human body dissipates excess heat are: 1) increased blood flow to the skin, which allows heat to be moved by convection from the interior of the body to the skin, and 2) secretion of sweat onto the skin's surface for subsequent evaporation.

When the external temperature is higher than the skin temperature, the only available heat release mechanism is through evaporation via sweat.

Heat loss through these two mechanisms increases in proportion to the rate of heat production and normally increases enough to balance metabolic heat production.

However, any factor that prevents evaporation, such as high humidity or reduced airflow (tight clothing, lack of breeze), will lead to increased body temperature and/or worsen chronic clinical conditions in more vulnerable individuals.

A key factor in the body's ability to cool through evaporation is the moisture gradient between the skin and the air. When there is sweat on the skin's surface and the ambient humidity is low (relative humidity of 20%), a relatively high rate of body cooling through evaporation is observed. In this case, the individual will withstand atmospheric temperatures of approximately 54.4°C.

In contrast, high relative humidity (80%) reduces the rate of sweat evaporation, significantly limiting the body's cooling capacity. In this situation, body temperature will increase whenever the ambient temperature exceeds approximately 34.4°C, causing dehydration.

In Portugal, the first situation occurs in inland regions, while the second occurs predominantly in coastal regions.

If a person is engaging in intense physical exertion, the critical ambient temperature level may be as low as 29.5°C to 32.2°C.

When air temperature and humidity levels exceed the optimal comfort range, health problems can begin to arise. The first effects are subjective in nature and relate to altered feelings of well-being, giving the individual the psychological sensation of being overheated. Consequently, the individual makes appropriate environmental adaptations to restore their sense of comfort – drinking water, seeking an air-conditioned room, changing clothing, among other things.

Another normal physiological process must be taken into account: acclimatization, which consists of developing a greater tolerance to conditions of increased temperature and humidity when there is progressive exposure over a period of 1 to 3 weeks.

However, exposure to high temperatures and humidity, particularly for several consecutive days, can cause heat-related illnesses such as cramps, exhaustion, and heat stroke (Figure 2).

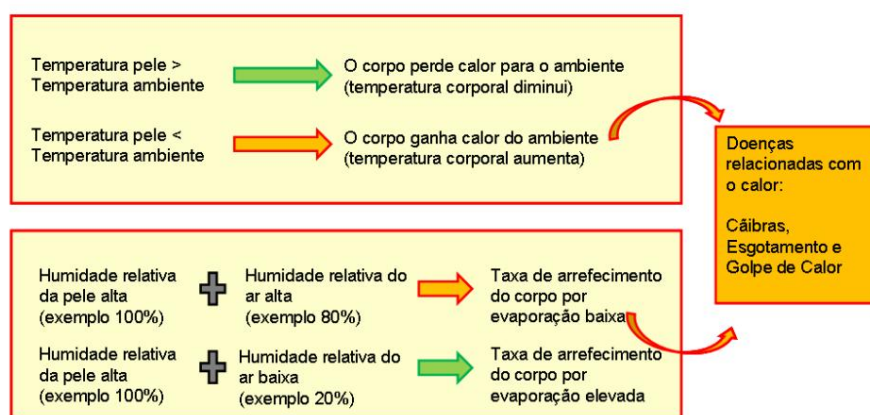


Figure No. 2 – Conditions that lead to the occurrence of heat-related illnesses

Fluid intake is the primary way to prevent adverse effects resulting from exposure to intense heat, and is essential even when the individual does not yet show signs of thirst. Annex IV provides a set of measures to prevent the effects of intense heat on health.

In addition to dehydration, intense heat can lead to the worsening of chronic diseases and the development of heat-related illnesses such as heat cramps, heat exhaustion, and heat stroke, the most serious condition caused by heat (Annex V).

In this context, and in order to minimize the effects of extreme temperatures on health, the Environmental and Occupational Health Division, integrated into the Directorate of Disease Prevention and Health Promotion Services of the DGS (Directorate-General of Health), has been promoting the preparation of Information Circulars/Guidelines on various aspects related to periods of intense heat and their effects on health (See Annex VI), available in the "Summer Special" area on the website www.dgs.pt.

3. OBJECTIVES OF THE CONTINGENCY PLAN

The overall objective of this Plan is to minimize the negative effects of intense heat on public health through effective risk assessment and the development of appropriate responses by competent health authorities, based on the provision of all relevant information and in collaboration with all entities involved.

This plan also has the following specific objectives:

- To enhance inter-institutional coordination;
- Improve the forecasting, warning, and response system;
- Define intervention guidelines/recommendations;
- To provide information to the general population, and to the most vulnerable groups in particular, about measures and procedures to adopt in situations of intense heat;
- Monitor morbidity and mortality resulting from potential heat waves.

4. INSTITUTIONAL ORGANIZATION AND ARTICULATION

For the implementation of the Contingency Plan, the financial resources are those derived from the annual allocations foreseen in the State Budget Law for the participating entities involved in the Plan.

With regard to human and organizational resources, the implementation of the actions included in the Contingency Plan requires a coordination of efforts from all stakeholders, intervening within their areas of expertise according to the alert level, in order to meet the necessary efficiency standards. Its operationalization requires a structure that optimizes efforts, developing and strengthening partnerships.

The institutional organization and coordination of the Heat Module of the Contingency Plan can be outlined according to the following figure.

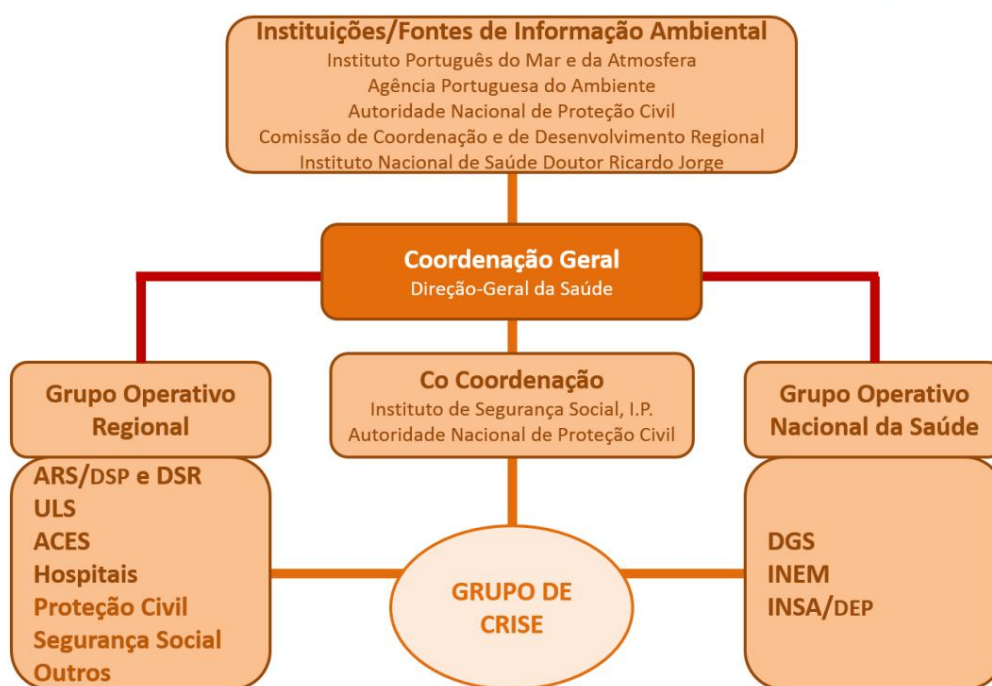


Figure No. 3 – Institutional Organization and Coordination

General Coordination

The coordination of the Heat Module of the Contingency Plan is carried out by the Directorate-General of Health, through the Environmental and Occupational Health Division, which is part of the Directorate of Disease Prevention and Health Promotion Services.

Its functions include coordinating, supervising, and evaluating the implementation of the Contingency Plan at the national level during its period of validity.

Coordination

The national co-coordination includes the National Civil Protection Authority and the Social Security Institute, IP.

Their functions include monitoring the progress of the Contingency Plan and promoting the dissemination of recommendations and measures adopted, together with the services they oversee. These entities play a fundamental coordinating role in the event of a crisis.

National Health Operational Group

The National Health Task Force is composed of members from the Directorate-General of Health, a representative from the National Institute of Medical Emergency (INEM), and a representative from the National Institute of Health Dr. Ricardo Jorge/Department of Epidemiology (INSA/DEP).

The Directorate-General of Health comprises the Directorate of Disease Prevention and Health Promotion Services through the Division of Environmental and Occupational Health, the Directorate of Information and Analysis Services through the Division of Epidemiology and Statistics, the Support Unit for the National Health Authority and Public Health Emergency Management (UESP), and the Support Unit for the National Health Service Call Center (Saúde 24).

This group's role is to operationalize the necessary means for the effective implementation of the Plan.

Institutions/Sources of Environmental Information

Environmental Institutions/Sources of Information include all those that provide baseline information for the daily risk assessment carried out by the Regional Operational Group, coordinated by the Regional Health Delegate in conjunction with the Health Delegates.

Coordinators, namely:

- Portuguese Institute of the Sea and the Atmosphere - observed and predicted daily extreme temperatures and ultraviolet radiation levels
- National Institute of Health Dr. Ricardo Jorge - Icarus Alert Index
- Environmental Agency - information on air quality
- National Civil Protection Authority - information on active fires.
- Regional Coordination and Development Commissions (CCDR) - ozone levels when excesses occur in their area of intervention.

Information regarding the ultraviolet index, air quality, and ozone excesses constitutes complementary criteria in risk assessment. In these cases, the competent authorities inform the population and the Health Delegates when levels likely to affect human health are reached. The Health Delegates then disseminate recommendations, through the media, to the population in the affected areas.

Regional Operational Groups

The Regional Health Administrations are responsible for preparing their respective Regional Contingency Plan and establishing the Regional Operational Group, providing the necessary resources for the implementation of the Plan.

According to the National Plan and the Regional Plan, the Local Health Units (ULS) and the Primary Care Units (ACES), through their Public Health Units, and the Hospitals will develop their specific Plans which they will submit to the Regional Health Administrations (ARS).

The Regional Operational Groups are based in their respective Regional Health Administrations/Primary Health Districts and coordinated by the Regional Health Delegates, who liaise with the Plan coordinators at the local level (the Coordinating Health Delegates).

Inter-institutional coordination should be foreseen with district civil protection and social security structures and with other entities, namely private social solidarity institutions, parishes, parish councils, municipal councils, firefighters, the Red Cross, the PSP (Public Security Police), the GNR (National Republican Guard) and others.

Crisis Group

Whenever the situation warrants it, the National Health Authority acts and coordinates the Group of Crisis, which consists of:

- Directorate-General of Health;
- National Civil Protection Authority;
- Social Security Institute, IP;
- National Institute of Health Dr. Ricardo Jorge, IP;
- National Institute of Medical Emergency;
- Portuguese Institute of the Sea and the Atmosphere;
- Regional Health Administrations, IP.

It may also be foreseen that Crisis Groups will be activated at the regional level, as established in the Regional Contingency Plans.

5. FORECASTING AND ALERT SYSTEM

Alert Levels and Criteria

The forecasting and alert system operates during the period in which the Contingency Plan is activated, with the Heat Module corresponding to the period from May 15th to September 30th, and may be extended depending on the weather conditions observed, at any time of year, before or after its activation period.

The Regional Operational Groups, coordinated by the Regional Health Delegates in conjunction with the Coordinating Health Delegates, are responsible for the daily assessment of risk at the regional and local levels, based on information provided by the Environmental and Occupational Health Division or from other regional/local sources and their knowledge of local specificities.

Three alert levels are defined: green (Level 0), yellow (Level 1), and red (Level 2).

The GREEN level corresponds to the baseline, meaning that the defined values are not reached to trigger the yellow alert.

A YELLOW alert is issued when a day occurs with observed maximum temperatures.

Temperatures equal to or greater than 32°C are expected, with two more days forecast to have temperatures equal to or greater than 32°C (except in the Alentejo region, where maximum temperatures equal to or greater than 35°C are considered).

A RED alert is issued when three days have recorded maximum temperatures equal to or greater than 35°C, and two more days are forecast with temperatures equal to or greater than 35°C (except in the Alentejo region, where maximum temperatures equal to or greater than 38°C are considered).

To establish these alert levels, the DGS (Directorate-General of Health) developed some generic and guiding criteria, which are presented in Annex I of this document. Similarly, the DGS suggests possible types of interventions, depending on the alert level, as detailed in Annex II.

Regional Health Administrations must include, in an objective manner, in their respective Regional Contingency Plans, the criteria used to define alert levels, as well as the respective action measures. The proposed criteria may be used and/or

Adapted according to the characteristics of each region, without prejudice to the use of other information considered relevant at the regional and/or local level.

Assignments and Information Flows

The State Health Secretariat, through the Division of Environmental and Occupational Health, makes available daily basic information, supplied by Environmental Institutions/Information Sources, so that the Regional Operational Groups can assess the risk. Information relating to the Icarus Alert Index is also made available daily by the National Institute of Health Dr. Ricardo Jorge IP. This information is entered into the electronic form available in the reserved area of the DGS website and sent via email.

Whenever justified, the Environmental and Occupational Health Division participates, via videoconference, in the **Technical Operational Briefing** held at the National Command of Relief Operations of the National Civil Protection Authority, with the Portuguese Institute of the Sea and the Atmosphere.

Following the information provided, each Regional Health Delegate and their respective The Working Group is defining the district alert levels (green, yellow, or red).

for the following day, which must be communicated to the Environmental and Occupational Health Division by 4 pm on the same day via *email* or by selecting the respective alert level for the districts in your region in the "daily information form" in the DGS reserved area.

The Environmental and Occupational Health Division, after receiving the district alert levels, makes the information available on the DGS website in the "Summer Special" area, generating a map with the district alerts and information on the municipal alerts of those districts that involve more than one Regional Health Administration.

In the event of a RED alert, the DGS (Directorate-General of Health) sends a communication to the LUSA news agency. The Regional Health Delegate is responsible for informing the Coordinating Health Delegates, who will then disseminate the information to the Local Health Units (ULS), Primary Care Units (ACES), and Hospitals not integrated into ULS, as well as to the Municipalities and other entities within their area of operation. They will also activate the measures deemed necessary, in accordance with the respective Contingency Plan, as well as disseminating recommendations to the population and issuing press releases at the regional level, if considered necessary.

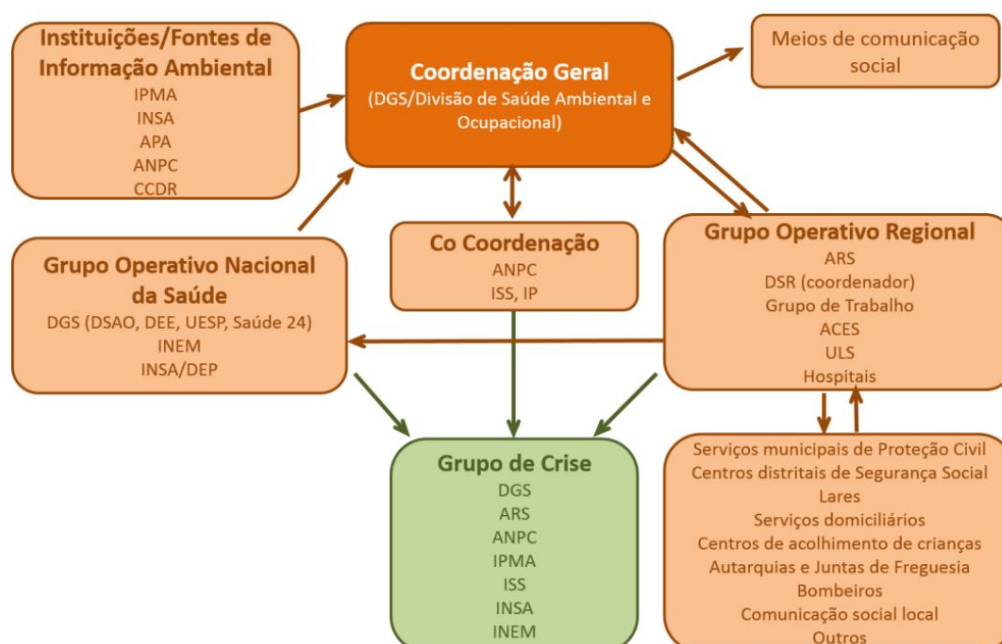


Figure No. 4 – Information Flows

The responsibilities assigned to each health entity represented in the Contingency Plan are specified in Annex III.

Bi-weekly reports and information

Every two weeks, the Environmental and Occupational Health Division, integrated into the Directorate of Disease Prevention and Health Promotion Services, prepares a monitoring report and assessment of the occurrences verified taking into account the analysis of the basic information and The monitoring indicators of the Plan. A statement based on this bi-weekly report is made available on the DGS website.

Monthly reports

Each Regional Operational Group, coordinated by the Regional Health Delegate, must prepare a monthly report that includes all occurrences related to the potential effects of heat, as well as the alert levels defined for that period and the respective measures.

Samples should be taken and sent to the Environmental and Occupational Health Division by the end of the first week of the following month.

Within their areas of responsibility, the entities represented in the National Health Operational Group must submit a monthly report to the Environmental and Occupational Health Division, evaluating what occurred during that period.

From all the information gathered, a monthly report is prepared and made available in the restricted area of the DGS website, for consultation by the Minister of Health, the Regional Health Administrations and the Regional Health Delegates.

Final report

At the end of the Plan's validity period, the Environmental and Occupational Health Division will prepare a final evaluation report on the implementation of the Heat Module of the Contingency Plan for subsequent submission to the Minister of Health by the end of December. For this purpose, the entities that... Those who coordinate and comprise the National Health Operational Group, and the Regional Operational Groups, must submit their respective final reports by the end of the first half of November.

The Epidemiology and Statistics Division, in conjunction with the National Institute of Health Dr. Ricardo Jorge, must prepare a final report on the evolution of mortality based on information from the Daily Mortality Surveillance System. In the event of a period of intense heat that gives rise to a meteorological heat wave, a report must be prepared assessing the impact on mortality for the period in question.

6. INFORMATION AND COMMUNICATION

Information for the general public is disseminated through:

- DGS website (www.dgs.pt), with daily updated information for the public, presenting, for this purpose, a map with district alerts and other relevant information for that district. Information is also available in the "Summer Special" section;
- Coordination with the media, which is a fundamental element in disseminating information to the public;
- Health 24 (808 24 24 24) for more personalized assistance in clarifying questions and support for situations related to heat.

Communication between the DGS (Directorate-General of Health) and the Health Services is done through the reserved area of the DGS website, fax, mobile phone or via the email address-calor@dgs.pt.

Incidents related to the effects of heat on public health, identified by local authorities and their respective municipal services, district social security centers, other institutions, or the general public, can be reported directly to the DGS (Directorate-General of Health) via the email address-calor@dgs.pt. They can also be reported locally to the Public Health Units of the Local Health Units/Health Center Groups.

7. HEALTH SERVICES RESPONSE

The Regional Health Administrations are responsible for developing and monitoring their respective Heat Module of the Regional Contingency Plan. The Plan is coordinated by the Regional Health Delegate, who liaises at the local level with the respective Health Delegates. Coordinators. These individuals promote the implementation of the Plan in partnership with others.

Institutions, with regard to identifying the most vulnerable population groups and to Identification and characterization of potential shelter sites.

The Regional Health Delegate, in coordination with the Coordinating Health Delegates and relevant partners, assesses the risk, defining a daily alert level. When deemed necessary, they may promote the dissemination of recommendations to the population and press releases at the regional level.

In the health sector, the Regional Health Administrations, the Local Health Units/Primary Care Units (ULS/ ACES), and hospitals not integrated into ULS ensure the necessary logistical resources so that appropriate measures can be taken at the defined alert level.

The Heat Module of the Regional Contingency Plans must be framed by the specific guidelines of the national plan, as well as by the complementary rules and guidelines prepared by the DGS (Directorate-General of Health).

The Heat Module of the Regional Contingency Plans must be sent to the Directorate-General of Health for their information by the end of May.

Regional Contingency Plans should include a set of general and specific actions to be taken; Local Health Units (ULS), Primary Care Units (ACES), and Hospitals not integrated into ULS must ensure the preparation of Specific Contingency Plans.

At the level of the Local Health Units (ULS) and the Primary Care Units (ACES), the Plans are coordinated by the Health Delegates. Coordinators, who report return information during yellow or red alert periods to the Regional Health Delegate, who then reports it to the respective Regional Health Secretariat.

This information is then entered into the appropriate form available in the reserved area of the DGS website.

When special situations occur, such as events with large gatherings of people, which are frequent during this period, namely sporting events, concerts, pilgrimages and processions, the Local Health Units/Primary Care Units and Hospitals not integrated into Local Health Units must ensure, in accordance with the Specific Contingency Plans, the measures and resources that are appropriate to address the situation. Whenever necessary, these Institutions and the Coordinating Health Delegate (who coordinates the Specific Contingency Plans) will coordinate with other entities, especially with Civil Protection and Social Security.

When shelter locations are activated, their location must be immediately transmitted by the Coordinating Health Delegate to the Regional Health Delegate, who will report it to... respective ARS.

The Regional Health Delegate will also communicate the information to the National Coordination of the Plan, through the Division of Environmental and Occupational Health, which, regardless of the dissemination at the local level, will make the information available on its website, through the Health 24 Service and the media.

8. PLAN MONITORING

Monitoring of the Heat Module of the Contingency Plan will be carried out based on the following aspects: demand for emergency services, demand for the 24-hour Health Service, demand for the National Institute of Medical Emergency, effects on mortality, incidents recorded by health authorities, and the actions taken within the scope of the Plan's implementation.

Monitoring the Demand for Emergency Services

Monitoring the demand for emergency services in hospitals and health center clusters is carried out using information recorded in the SINUS and SONHO computer applications, managed by the Central Administration of the Health System.

Monitoring the Demand for the Health 24 Service

The Health 24 Service activates the "summer module" during the period from May 15th to September 30th. Periodic mechanisms are created to ensure the flow of information between the DGS (Directorate-General of Health) and the Line Agents, and for feedback, activating procedures that facilitate access to the Internet and the professional email of each Line Agent. Whenever necessary, information from the Line Agents will be forwarded to the Regional Operational Groups through the DGS.

Monitoring the Demand for INEM Services

The National Institute of Medical Emergency is responsible for providing immediate assistance to people in emergency situations and ensuring their transport to the appropriate healthcare facility.

You must provide the DGS (Directorate-General of Health) with the total daily number of service requests by district, the number of calls per algorithm, and the number of activations nationwide.

The DGS (Directorate-General of Health) transmits all this information to the Regional Operational Groups.

Monitoring the Effects on Mortality

The monitoring of the daily evolution of mortality associated with periods of intense heat is ensured by the Epidemiology Department of the National Institute of Health Dr. Ricardo Jorge, through the Daily Mortality Surveillance system and the definition of the Icarus Alert Index.

The Epidemiology and Statistics Division must monitor the evolution of mortality information and, in collaboration with the National Institute of Health Dr. Ricardo Jorge, prepare a report on preliminary mortality estimates at the end of the Plan's validity period and in the event of a meteorological heat wave with a significant impact on mortality.

The Coordinating Health Delegates must report information on the occurrence of deaths directly related to heat to the Regional Health Delegates, and these, in turn, must report to the Regional Health Delegates. their respective Regional Health Administrations. These must communicate any information they become aware of to the Environmental and Occupational Health Division.

Monitoring of the Measures Taken

Monitoring the measures taken by the Local Health Units/Primary Care Units and by hospitals not integrated into Local Health Units, It is based on the feedback information communicated by the Coordinating Health Delegate to the Regional Health Delegate and the respective Regional Health Administration, being subsequently entered into the appropriate form available in the reserved area of the DGS website, via email (calor@dgs.pt) or through other means of communication that are deemed appropriate. In this form, in addition to the type of intervention performed, occurrences related to the effects of heat on health may be reported.

9. DISSEMINATION

The DGS (Directorate-General of Health) has been developing several Information Guidelines/Circulars (see Annex IV) with general recommendations on the effects of intense heat on health and with specific recommendations taking into account more vulnerable groups or those with particular characteristics.

Two guidelines/circulars were prepared, one for the general population and another on the characteristics of appropriate clothing for periods of high temperatures. Guidelines/circulars with recommendations for children, the elderly, homeless people, outdoor workers, tourists, people with kidney failure, pregnant women, athletes, and people with diabetes are also available.

These guidelines/circulars are an initiative of the general coordination of the National Plan and have been technically reviewed by experts in various fields, including Pediatricians, Geriatricians, specialists in General and Family Medicine, Public Health and Environmental Health.

Disseminating information to the general public and the most vulnerable groups should be promoted by each region at the beginning of the implementation of the Contingency Plan and continuously throughout the entire period of validity of the Plan.

Therefore, the Regional Health Administrations and the Local Health Units/Primary Care Units should include this in their Plans. Awareness-raising activities for healthcare professionals, professionals providing care to vulnerable groups, and other interested parties.

The information produced is made available online on the DGS website and the websites of the various Health Regions. Any entity wishing to print leaflets and/or posters for dissemination can download them *directly* from the website www.dgs.pt.

For the purpose of disseminating information to the general public, it is also planned to produce bi-weekly reports detailing the evolution of environmental conditions and heat-related events.

In situations where a yellow or red alert level is triggered, information is prominently displayed on the DGS website, and in the case of a red alert level, a *pop-up* map of the country is activated showing district and municipal alert levels, which, because they belong to different health regions, may differ from the district alert, and a statement is issued to the LUSA news agency.

If deemed necessary, the Crisis Group, once activated, may decide on other means of dissemination.

10. INVESTIGATION

Research into the effects of high temperatures on health is essential as a tool to support the development of strategies and intervention plans, as well as to fill knowledge gaps and consolidate existing knowledge.

Projects involving inter-institutional collaboration at the national and international levels, particularly with the World Health Organization, are considered a priority in order to integrate the... best available knowledge about the effects of heat on human health.

11. EVALUATION OF PLAN IMPLEMENTATION

The evaluation of the Heat Module of the Contingency Plan will be carried out through an Implementation Evaluation Report, taking into account regional and national levels.

The regional assessment will be based on the monthly reports and the final report, prepared by each Regional Health Delegate and by the Coordinating Health Delegates and their respective Working Groups. The Final Report for each Health Region must be sent to the DGS (Directorate-General of Health) by the end of [the deadline] from the first half of November.

The national-level evaluation will be carried out through the analysis of monthly reports, taking into account the measures implemented throughout the Plan and analyzing the following indicators:

- Identifying periods of intense heat and the frequency with which the temperatures that characterize them occur;
- Characterization of heat-related events;
- Characterization of the demand for Saúde 24 services;
- Analysis of the demand for emergency services;
- Analysis of the demand for INEM services;
- Evaluation of the measures taken, in particular, through feedback information;
- Analysis of the relationship between recorded and expected mortality.

The **final evaluation report** on the implementation of the Heat Module of the Contingency Plan is the responsibility of the Directorate-General for Health, through the Division of Environmental and Occupational Health, with the contribution of all entities involved in this Plan.

This report will be brought to the attention of the Minister of Health by the end of December, and will subsequently be made available on the DGS website for consultation by all entities and the general public.

12. Long-Term Planning

The Heat Module of the Contingency Plan for Adverse Extreme Temperatures is designed from a short-term perspective, considering the evolving weather conditions during the summer season. However, the best way to prevent the effects of intense heat on human health is through the preparation of the spaces in which we live, both public and private; that is, acting at the level of land use planning, urban planning, and conditions inside homes.

In this regard, the Directorate-General of Health has been developing guidance tools aimed at promoting the adoption of best practices, taking into account the broad concept of human health.

Therefore, it is strongly recommended that you consult the following documents available on the website www.dgs.pt:

- Fact Sheets on Housing and Health;
- Integration and Appreciation of the Human Health Component in Municipal Plans of Spatial Planning (Information Circular No. 36/DA of 09.10.2009).

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APPENDICES

ANNEX I – CRITERIA FOR DEFINING ALERT LEVELS

According to the definition of the Heat Wave Duration Index (HWDI) by the World Meteorological Organization (WCDMP-No. 47, WMO-TD No. 1071), a heat wave is considered to occur when, over a period of at least six consecutive days, the maximum air temperatures are 5°C higher than the average maximum temperatures during the reference period (IM).

It should be noted, however, that this definition is more related to the study and analysis of climate variability than to its impacts on public health.

Regarding the effects of heat on health, the DGS (Directorate-General of Health) has established some generic criteria that allow for the definition of alert levels, which should be considered collectively and not individually. It should be noted that the criteria outlined here are only generic guidelines, and the Regional Operational Groups should develop their own criteria based on local geodemographic specificities.

GREEN ALERT – corresponds to the baseline, at which the values of reference foreseen in the criteria of the yellow level.

YELLOW ALERT - the following criteria are proposed:

CRITERION 1 – Icarus Alert Index

A positive Icarus Alert Index (Icarus Alert Index) of less than 1 indicates a non-significant effect on mortality, but likely effects on morbidity.

Criterion	Meaning		Application
Icarus Alert Index	0,01 a 0,99	No significant effect on mortality.	The highest value among the three regional daily values (D; D+1; D+2) = “maximum”

CRITERION 2 – Maximum temperatures

The yellow alert will be activated after one day with a recorded maximum temperature and two days with forecasts of temperatures equal to or greater than 32°C and less than 35°C, for the months of May and June.

For the remaining months, one day with a maximum observed temperature equal to or greater than 32°C and less than 35°C, and two days with predicted temperatures equal to or greater than 32°C are considered for all regions except Alentejo. For Sergipe, one day with a maximum observed temperature and two days with predicted temperatures equal to or greater than 35°C and less than 38°C are considered.

Period	Maximum Temperatures by Region	Application
May - June	$\geq 32^{\circ}\text{C}$ and $< 35^{\circ}\text{C}$ (all regions)	1 day observed temperature + 2 days predicted temperatures
July, August and September	$\geq 32^{\circ}\text{C}$ and $< 35^{\circ}\text{C}$ (except Alentejo)	
	$\geq 35^{\circ}\text{C}$ and $< 38^{\circ}\text{C}$ (Alentejo)	

CRITERION 3 – Minimum temperatures

The yellow alert will be activated if minimum temperatures exceed comfort temperatures (21°C – 23°C), preventing the human body from cooling down at night; that is, after two days of observed minimum temperatures and two days with forecasts of minimum temperatures equal to or greater than 24°C.

Period	Minimum Temperatures by Region	Application
May - September	$\geq 24^{\circ}\text{C}$ and $< 26^{\circ}\text{C}$ (all regions)	2 days of observed temperatures + 2 days of predicted temperatures

CRITERION 4 – Fires

The yellow alert will be activated after a forecast of two days in which the temperature values of criterion 2 are reached, in the event of a day of fire spread with a forecast of its continuation into the following day and with pollutants being transported by the wind to densely populated areas.

Period	Maximum Temperatures by Region	Application
May - June	$\geq 32^{\circ}\text{C}$ and $< 35^{\circ}\text{C}$ (all regions)	Fire + 2 days of predicted temperatures
July, August and September	$\geq 32^{\circ}\text{C}$ and $< 35^{\circ}\text{C}$ (except Alentejo)	
	$\geq 35^{\circ}\text{C}$ and $< 38^{\circ}\text{C}$ (Alentejo)	

CRITERION 5 – Other factors

Other factors should also be considered, such as: ozone level excesses, ultraviolet radiation levels, local events, weather warnings, and the Universal Thermal Climate Index. UTCI (from the Portuguese Institute of the Sea and the Atmosphere).

 **RED ALERT** – the following criteria are proposed:

CRITERION 1 – Icarus Alert Index

An Icarus Alert Index equal to or greater than 1.0 indicates probable effects and potentially serious expected consequences on health and mortality.

Criterion	Meaning		Application
Icarus Alert Index	$\geq 1,0$	Likely effects and expected serious consequences on health and mortality.	The highest value among the three regional daily values (D; D+1; D+2) = “maximum”

CRITERION 2 – Maximum temperatures

The red alert will be activated after three days of observed maximum temperatures and two days with forecasts of temperatures equal to or greater than 35°C, for the months of May and June.

For the remaining months, three days with observed maximum temperatures equal to or greater than 35°C and two days with predicted temperatures equal to or greater than 35°C are considered for all regions, except Alentejo. For Sergipe, three days with observed maximum temperatures and two days with predicted temperatures equal to or greater than 38°C are considered.

Period	Maximum Temperatures by Region	Application
May – June	$\geq 35^{\circ}\text{C}$ (all regions)	3 days of observed temperatures + 2 days of predicted temperatures
July, August and September	$\geq 35^{\circ}\text{C}$ (except Alentejo)	
	$\geq 38^{\circ}\text{C}$ (Alentejo)	

If maximum observed temperature information is missing for one or two days, limiting the application of this criterion, it may be considered to combine it with criterion 4 (minimum temperatures), also using the maximum temperatures predicted for the days in which the observed temperatures are missing.

CRITERION 3 - Sudden temperature rise equal to or greater than 6°C

Following a sharp rise in maximum temperature equal to or greater than 6°C , and reaching the values of Criterion 2, the red alert will be activated with two days of observed temperatures and two days of predicted temperatures.

Period	Maximum Temperatures by Region	Application
May – June	$\geq 35^{\circ}\text{C}$ (all regions)	2 days of observed temperatures + 2 days of predicted temperatures
July, August and September	$\geq 35^{\circ}\text{C}$ (except Alentejo)	
	$\geq 38^{\circ}\text{C}$ (Alentejo)	

CRITERION 4 – Minimum temperatures

The red alert will be activated if minimum temperatures are so high that they do not allow the human body to cool down at night, that is, after two days of observed minimum temperatures and two days with forecasts of minimum temperatures equal to or greater than 26°C .

Period	Minimum Temperatures by Region	Application
May - September	$\geq 26^{\circ}\text{C}$ (all regions)	2 days of observed temperatures + 2 days forecast temperatures

CRITERION 5 – Fires

The red alert will be activated after two days of observed maximum temperatures and a forecast of two days in which the temperature values of criterion 2 are reached, in the event of a day of fire spread with a forecast of its continuation into the following day and with pollutants being transported by the wind to densely populated areas.

Period	Maximum Temperatures by Region	Application
May – June	$\geq 35^{\circ}\text{C}$ (all regions)	Fire + 2 days of observed temperatures + 2 days of predicted temperatures
July, August and September	$\geq 35^{\circ}\text{C}$ (except Alentejo)	
	$\geq 38^{\circ}\text{C}$ (Alentejo)	

CRITERION 6 – Other factors

Other factors should also be considered, such as: ozone level excesses, ultraviolet radiation levels, local events, weather warnings, and the Universal Thermal Climate Index - UTCI (from the Portuguese Institute of the Sea and the Atmosphere).

ANNEX II – ALERT LEVELS AND TYPES OF INTERVENTIONS

Alert level	Situation	Measures/Interventions
Level 1 Green Alert	Normal temperatures for this time of year of the year.	- Maintain a state of vigilance - To ensure the maintenance of general measures.
Level 2 Yellow Alert	High temperatures can have health effects.	- Dissemination of information and recommendations to the population, to Competent health authorities and other institutional sectors, including the media; - To strengthen the response capacity of healthcare providers and other units.
Level 3 Red Alert	Very high temperatures can cause serious problems for the health.	- Dissemination of information and recommendations to the population, to Competent Health Authorities and other institutional sectors, including the media; - Coordinate with health entities and with the partner entities included in the Operational Groups Regional; - Coordinate with the relevant entities emergency measures to facilitate transportation to shelter locations; - To ensure support for the most vulnerable groups – institutionalized elderly people, children, and people living in isolation or with reduced mobility; - To ensure responsiveness of healthcare provider units.

ANNEX III - RESPONSIBILITIES BY ORGANIZATIONS AND SERVICES OF THE MINISTRY OF HEALTH

Entity	Responsibilities
Environmental Health Division and Occupational Health (DSAO)	- To provide all the basic information necessary to the Regional Operational Groups for the daily risk assessment. This information includes: extreme temperatures (observed and predicted), Icarus Alert Index, ozone level exceedances, UV radiation levels and other relevant elements; - Update the content of the website www.dgs.pt Generating a map with district alerts and municipal information by area of intervention of local health services for districts that involve more than one Regional Health Administration (ARS); - Prepare bi-weekly, monthly, and final evaluation reports on the implementation of the Contingency Plan; - Prepare bi-weekly press releases and Notified to LUSA when justified; - Participate in research projects and develop recommendations for vulnerable groups and other heat-related situations; - To disseminate and manage the information from the email address calor@dgs.pt.
Epidemiology Division and Statistics (DEE)	- Coordinate with the National Institute of Health Doctor Ricardo Jorge, for joint analysis of mortality estimates; - To prepare, based on mortality data, reports with preliminary estimates of excess mortality at the end of the surveillance period and in the event of a meteorological heat wave with a significant impact on mortality.
Support Unit for National Authority and to Emergency Management in Public Health (UESP)	- Collaborate with the Environmental Health Division and Occupational health monitoring of the Support System Public Health Emergencies (seeking healthcare in emergency services); - To ensure the necessary mechanisms for the Communication of the measures to be adopted in a public health crisis situation; - To ensure the functioning of the necessary communication platforms.

Entity		Responsibilities
	Support Unit for Customer Service Center National Health Service (Health 24)	- To provide clarification and information to healthcare professionals regarding situations of intense heat and their effects on health, referral locations, and alert levels; - To advise citizens on the most appropriate measures. appropriate measures to take to avoid heat-related health problems and referral locations; - Provide information about shelter locations for collective use.
	Regional Administration of Health (ARS)	- To ensure the logistical resources necessary for the implementation of the Plan at the regional level; - To establish the Regional Operational Group; - Coordinate and develop Contingency Plans Regional; - To create the conditions for mobilizing an appropriate response; - To ensure the response of health services.
	Regional Health Delegate (DSR)	- To coordinate with various regional or district-level entities in order to align competencies and respective interventions; - To conduct daily risk assessments in conjunction with the Health Delegates Coordinators will define the daily district and/or municipal alert levels; - Monitor the implementation of the measures considered. necessary depending on the alert level defined. - Ensure that feedback information is sent to the DGS (Directorate-General of Health). through the Division of Environmental and Occupational Health; - Disseminate recommendations and make announcements of Regional press coverage, if deemed necessary; - Prepare monthly and final evaluation reports on the implementation of the Contingency Plan at the Health Region level.

Entity	Responsibilities
Health Delegate Coordinator (DSC) and partners	- Coordinate with the Regional Health Delegates and their respective Regional Health Management offices to develop specific contingency plans and define daily alerts; - Identify the most vulnerable groups; - Identify and characterize shelter locations; - Implement the action measures according to the alert level; - Ensure that feedback information is sent to the Regional Health Delegate.
National Institute of Health Doctor Ricardo Jorge/ Department of Epidemiology (DEP)	- To develop the Icarus Alert Index and its respective bulletin; - Prepare the Daily Mortality Surveillance report; - To contribute to the preparation of the final report with estimates of excess mortality attributed to heat in coordination with the DGS (Directorate-General of Health) and in the event of a meteorological heat wave with a significant impact on mortality; - Development of investigative actions.
National Institute of Medical Emergency (INEM)	- To ensure an appropriate response and transport for sick people in situations of intense heat; - Monitor the occurrences of requests for your services by district.

ANNEX IV - PRECAUTIONS TO TAKE DURING PERIODS OF INTENSE HEAT

In general, and to prevent the effects of intense heat, the following measures are recommended:

• Increase your intake of water, or natural fruit juices without added sugar, even without being thirsty.

People who suffer from chronic illness, or who are on a low-salt or fluid-restricted diet, should consult their doctor or contact the...

Health Line 24: 808 24 24 24.

• Avoid alcoholic beverages and drinks with high sugar content.

Newborns, children, the elderly, and the sick may not feel or show thirst, making them particularly vulnerable – offer them water and be attentive and vigilant.

Light and more frequent meals should be eaten. Heavy meals should be avoided.
very spicy.

Spending two to three hours a day in a cool or air-conditioned environment can prevent the harmful consequences of heat, particularly for children, the elderly, or people with chronic illnesses. If you don't have air conditioning, visit shopping malls, cinemas, museums, or other cool places. Avoid sudden temperature changes. Find out about the availability of air-conditioned shelters near you.

• During the hottest part of the day, take a lukewarm or cold shower. However, avoid sudden changes in temperature (an ice-cold shower immediately after being very hot can cause hypothermia, especially in elderly people or children).

• Avoid direct sun exposure, especially between 11 am and 5 pm. Whenever you are exposed to the sun or outdoors, use sunscreen with a high protection factor (SPF 30 or higher) and reapply it whenever you are exposed to the sun (every 2 hours) and if you are wet or have sweated a lot. When you return from the beach or pool, reapply sunscreen, especially during the hottest hours and when ultraviolet radiation is high.

When walking outdoors, wear clothing that avoids direct skin exposure to the sun, particularly during peak sun hours. Wear a hat, preferably with a wide brim, and sunglasses that offer protection against UVA and UVB radiation.

• Avoid staying in vehicles exposed to the sun, especially during the hottest periods, particularly in traffic jams and parking lots. If the car does not have air conditioning, do not close the windows completely. Bring enough water or natural fruit juices without added sugar for the trip and stop to drink them. Whenever possible, travel at night.

• Never leave children, sick people, or elderly people inside vehicles exposed to the sun.

• Whenever possible, reduce physical exertion and rest frequently in safe places.

Shady, cool, and well-ventilated areas. Avoid activities that require physical exertion.

• Wear loose, light, and cool clothing, preferably made of cotton and in accordance with the Information Circular No. 23/DA of 02/07/2009.

• Use less bedding, especially when dealing with babies and patients.
bedridden

• Prevent heat from entering homes. Close blinds or shutters to keep air circulating inside the house. In the evening, when the outside temperature is lower than the inside temperature, create drafts, being mindful of the harmful effects this can have.

• Do not hesitate to ask a family member or neighbor for help if you feel unwell.
heat.

• Regularly check on the health status of isolated, elderly, frail, or dependent people living near you and help them protect themselves from the heat.

Elderly people should not go to the beach on very hot days. Children under six months should not be exposed to the sun, and direct sun exposure should be avoided for children under three years old. Solar radiation can cause sunburn, even under a parasol; seawater and beach sand also reflect solar rays, and being in the water does not prevent sunburn in exposed areas. Sunburn reduces the skin's ability to cool down.

ANNEX V – SERIOUS EFFECTS OF INTENSE HEAT ON HEALTH – SYMPTOMS AND MEASURES OF PREVENTION

Our bodies strive to maintain a constant internal body temperature of 37°C over time. During periods of intense heat, the body produces sweat, which is the primary way it cools down as the sweat evaporates.

When air humidity levels increase, sweat cannot evaporate as quickly as it should. Sweat evaporation stops completely when the relative humidity reaches 90%. Under these circumstances, body temperature rises, and the resulting increase in sweat production can lead to excessive dehydration, potentially causing irreversible damage to the brain or other organs, or even death.

In extreme situations of exposure to intense heat, particularly over several consecutive days, heat-related illnesses may arise, such as heat cramps, heat exhaustion, and heat stroke, conditions which, due to their severity, may require emergency medical care.

Heatstroke

This situation occurs when the body's temperature control system stops working, ceasing to produce sweat to cool the body. Body temperature can reach 39°C in 10-15 minutes, causing brain damage or even death if the individual is not treated quickly.

Symptoms

Symptoms include high fever, red, hot, dry skin with no sweating, rapid and strong pulse, headache, nausea, dizziness, confusion, and partial or total loss of consciousness.

What to do?

Immediately call a doctor or dial the emergency number 112, following these procedures until their arrival.

- Move the individual to a cool place or to an air-conditioned room;
- Refresh the individual by applying damp towels or spraying them with cold water.
body;
- Ventilate the individual by vigorously agitating the air or using a fan;

If the baby is not conscious, do not give them any liquids.

Heatstroke requires immediate medical attention, as delayed treatment can result in complications affecting the brain, kidneys, and heart.

Heat exhaustion

This results from an alteration in hydroelectrolytic metabolism caused by excessive loss of water and electrolytes through sweating. This situation can be especially serious in elderly people and people with high blood pressure.

Symptoms

Symptoms include intense thirst, profuse sweating, paleness, muscle cramps, fatigue and weakness, headache, nausea and vomiting, and fainting. Body temperature may be normal, below normal, or slightly above normal. The pulse becomes thready, alternating between weak and rapid, and breathing becomes rapid and shallow.

What to do?

Immediately call a doctor or dial the emergency number 112, following these procedures until their arrival.

- Move the individual to a cool place or to an air-conditioned room;
- Refresh the individual by applying damp towels or spraying them with cold water.
body;
- Lay the individual down and raise their legs;
- Give them natural fruit juices without added sugar and/or drinks containing electrolytes.
(sports drinks), if you are conscious. _____

Heat cramps

Muscle cramps can result from simple exposure to intense heat, excessive sweating after periods of intense physical exercise, and inadequate hydration with water alone without replacing the electrolytes lost through perspiration.

Although less serious than the previous ones, this situation may also require medical treatment. Cramps are especially dangerous in people with heart problems or on low-salt diets.

Symptoms

They manifest as painful muscle spasms in the abdomen and extremities (legs and arms), caused by the loss of salts and electrolytes.

What to do?

- Stop exercising, if necessary, and rest in a cool, quiet place;
- Stretch the muscles and massage gently;
- Drink natural fruit juices without added sugar and/or drinks containing electrolytes (sports drinks);
- Seek medical help if the cramps persist.

To avoid all these situations caused by exposure to intense heat, protect yourself from sun exposure and seek cool or air-conditioned places during the hottest period, especially if you are accompanied by young children, elderly people, or people with chronic illnesses.

ANNEX VI – INFORMATION CIRCULARS/GUIDELINES

Published

General recommendations for the population

Information Circular No. 24/DA, dated 09/07/2009

Recommendations on appropriate clothing for periods of high temperatures.

Information Circular No. 23/DA, dated 02/07/2009

Recommendations for nurseries and kindergartens

Information Circular No. 30/DSAO, dated July 21, 2010

Recommendations for elderly care facilities

Information Circular No. 31/DSAO, dated July 21, 2010

Recommendations for those responsible for supporting homeless people.

Information Circular No. 32/DSAO, dated July 21, 2010

Recommendations for workers abroad

Information Circular No. 33/DSAO, dated July 21, 2010

Recommendations for tourists

Guidance Note No. 14/2011, dated 16/05/2011

Guidance Note No. 15/2011, dated 16/05/2011 (English version)

Recommendations for kidney failure

Information Circular No. 29/DSAO, dated July 21, 2010

Recommendations for pregnant women

Information Circular No. 25/DA/DSR, dated July 20, 2009

Recommendations for athletes

Information Circular No. 29/DA, dated 07/08/2009

Prevention of health risks associated with smoke from fires.

Guidance Note No. 16/2011, dated 16/05/2011

Recommendations for people with diabetes

Guidance Note No. 12/2012, dated 31/08/2012



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